Probability Theory and Statistics, Tutorial 5

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Continuous Random Variable

Two key properties:

- Possible values comprise either:
  - A single interval on the number line, OR
  - A union of disjoint intervals
- For any number c, $P(X = c) = 0$

Definition

Let X be a continuous rv. A probability density function (pdf) of X is a function $f(x)$ such that for any two numbers a and b with $a \leq b$:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Definition

The cumulative distribution function $F(x)$ for a continuous rv X is defined for every number x by:

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(y)dy$$

Key Property

If X is a continuous rv with pdf $f(x)$ and cdf $F(x)$, then at every x where the derivative exists:

$$F'(x) = f(x)$$

Discrete random variable	Continuous random variable
Countable set (finite or countably infinite): e.g., $\{0, 1, 2, \dots\}$	Uncountable set (typically intervals): e.g., $\mathbb{R}$, $[0, 1]$
PMF $p_X(x) = \Pr(X = x)$	**PDF** $f_X(x) \geq 0$ (density, not a probability)
Can be positive: $\Pr(X = x_0) = p_X(x_0)$	Always 0: $\Pr(X = x_0) = 0$
$\Pr(X \in A) = \sum_{x \in A} p_X(x)$	$\Pr(X \in A) = \int_A f_X(x)\, dx$
Often written via sums: $\Pr(a \leq X \leq b) = \sum_{x \in [a,b]} p_X(x)$	$\Pr(a \leq X \leq b) = \int_a^b f_X(x)\, dx$
$F_X(x) = \Pr(X \leq x) = \sum_{t \leq x} p_X(t)$ (step function)	$F_X(x) = \Pr(X \leq x) = \int_{-\infty}^x f_X(t)\, dt$ (usually continuous)

Definition

The expected or mean value of a continuous rv X with pdf $f(x)$ is:

$$\mu_X = E(X) = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

Definition

The variance of a continuous rv X with pdf $f(x)$ and mean μ is:

$$\sigma_X^2 = V(X) = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot f(x)dx = E[(X-\mu)^2]$$

The standard deviation is $\sigma_X = \sqrt{V(X)}$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

General Property

If X is continuous with pdf $f(x)$ and $h(X)$ is any function of X:

$$E[h(X)] = \mu_{h(X)} = \int_{-\infty}^{\infty} h(x) \cdot f(x)dx$$

Definition

A continuous rv X is said to have a uniform distribution on the interval $[A, B]$ if the pdf of X is:

$$f(x; A, B) = \begin{cases} \frac{1}{B-A} & A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$

Exercise 4

A college professor never finishes his lecture before the end of the hour and always finishes within 2 minutes after the hour. Let X be the time that elapses between the end of the hour and the end of the lecture, which pdf

$$f(x) = \begin{cases} kx^2, & \text{if } 0 \leqslant x \leqslant 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) Identify the value of k, and then derive the cdf for X.

(b) Find the probability that the lecture ends within 1 minute of the end of the hour.

(c) What is the probability that the lecture continues beyond the hour for between 60 and 90 seconds?

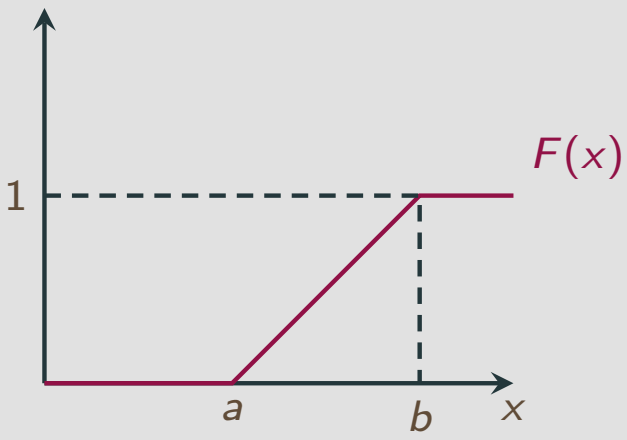
(d) What is the probability that the lecture continues for at least 90 seconds beyond the end of the hour?

(e) If the probability that the lecture ends within m minutes of the end of the hour equals 0.5, find the value of m.

(f) Calculate $\mathbf{E}(X)$ and $V(X)$.

Exercise 5

The graph of the cdf of a random variable X is shown in the figure below. The graph consists of three phases, each being a segment or a ray.



(a) Find the expressions of the cdf and pdf of X, respectively.

(b) Which type of distribution does X follow?

(c) Calculate the expected value and variance of X.

For the variance, note that $V(X) = \mathbf{E}(X^2) - (\mathbf{E}(X))^2$. Note that

$$\mathbf{E}(X^2) = \int_a^b x^2 \cdot \frac{1}{b-a}\,\mathrm{d}x = \frac{1}{b-a} \cdot \frac{x^3}{3}\bigg|_a^b = \frac{a^2+ab+b^2}{3}.$$

Thus,

$$V(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}.$$

Definition

A continuous random variable X has a normal distribution with parameters μ and σ (or μ and σ^2), where $-\infty < \mu < \infty$ and $\sigma > 0$, if its pdf is:

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} \qquad -\infty < x < \infty$$

- Notation: $X \sim N(\mu, \sigma^2)$
- Parameters:
  - μ = mean (location parameter)
  - σ = standard deviation (scale parameter)

Standardization

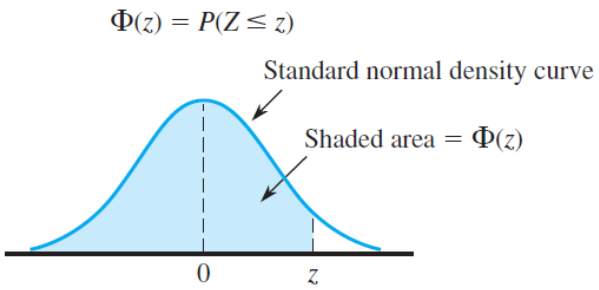
If $X \sim N(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt\,.$$

- Allows use of standard normal table for any normal distribution
- Probability calculations:

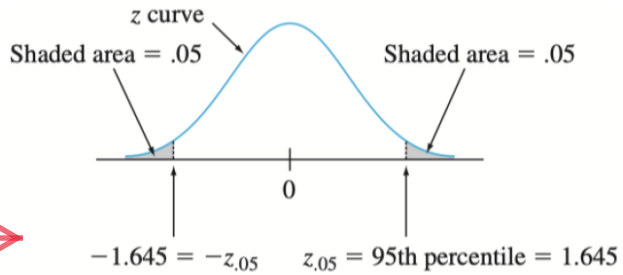
$$P(a \leq X \leq b) = P(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma})$$



z_α Critical Values

Definition

z_α denotes the value on the z axis for which α of the area under the z curve lies to the right of z_α



Finding Percentiles

For any p between 0 and 1, we can find the value z_p such that:

$$P(Z \leq z_p) = p$$

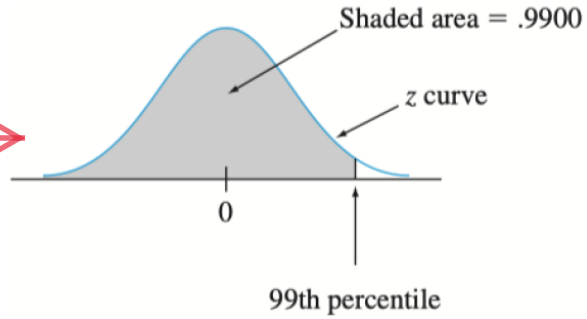


Table A.3 Standard Normal Curve Areas *(cont.)* $\Phi(z) = P(Z \leq z)$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9278	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936

Exercise 1

Suppose a random variable X follows standard normal distribution.

(a) Find the value of $P(-1 < X < 2)$, $P(|X| \geqslant 1)$, $P(|X| < 2)$, respectively.

(b) If the percentile $z_p = \alpha > 0$, express $P(X \geqslant \alpha)$, $P(|X| < \alpha)$, and $P(X < -\alpha)$ in terms of p.

(c) Determine the value of c for each of the following cases:

  (i) $P(0 \leqslant X \leqslant c) = 0.291$;

  (ii) $P(|X| \leqslant c) = 0.668$.

(d) Determine the critical value z_α for $\alpha = 0.09$.

Exercise 2

Extensive data supports the assumption that for babies born in the "normal" range of 37-43 weeks gestational age, birth weight is normally distributed with mean 3432 g and standard deviation 482 g.

(a) What is the probability that the birth weight of a randomly selected baby of this type exceeds 4000 g? Is between 3000 g and 4000 g? Is either less than 2000 g or more than 5000 g?

(b) How would you characterize the most extreme 2% of all birth weights?

(c) Suppose 1 lb $= a$ g. If we now measure birth weight using lb rather than g, what is the new distribution of birth weight?

Definition

A random variable X has an exponential distribution with parameter $\lambda > 0$ if its pdf is:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Mean: $E(X) = \frac{1}{\lambda}$
- Variance: $V(X) = \frac{1}{\lambda^2}$
- Standard Deviation: $\sigma = \frac{1}{\lambda}$

Cumulative Distribution Function

$$F(x;\lambda) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Key Property

For any $t, t_0 \geq 0$:

$$P(X \geq t + t_0 | X \geq t_0) = P(X \geq t)$$

- Interpretation:
  - Future lifetime is independent of current age
  - No "memory" of how long it has been working
  - Unique to exponential distribution

Exercise 3

Let random variable X be the time between two successive arrivals at the drive-up window of a local bank. Suppose X is distributed exponentially with the parameter $\lambda = 1$.

(a) What is the expected time between two successive arrivals?

(b) What is the standard deviation of the time between successive arrivals?

(c) Calculate $P(X \leqslant 4)$.

(d) Calculate $P(2 \leqslant X \leqslant 5)$.

(e) Calculate $P(X \geqslant 3 \mid X \geqslant 1)$.

(a) $\mathbf{E}[X] = \dfrac{1}{\lambda} = \dfrac{1}{1} = 1.$

(b) $\sigma_X = \dfrac{1}{\lambda} = \dfrac{1}{1} = 1.$

(c) $P(X \leqslant 4) = F(4) = 1 - \mathrm{e}^{-4} = 0.9817.$

(d) $P(2 \leqslant X \leqslant 5) = F(5) - F(2) = (1 - \mathrm{e}^{-5}) - (1 - \mathrm{e}^{-2}) = \mathrm{e}^{-2} - \mathrm{e}^{-5} = 0.1286.$

(e) Employing the "memoryless" property of exponential distributions, we have

$$P(X \geqslant 3 \mid X \geqslant 1) = P(X \geqslant 2 + 1 \mid X \geqslant 1) = P(X \geqslant 2) = 1 - P(X \leqslant 2) = 1 - (1 - \mathrm{e}^{-2}) = 0.1353.$$

Exercise 4

Starting from a certain time $t_0 = 0$, the number of customers who will visit a shop is a *Poisson process* with rate $\alpha > 0$ per hour. Let X_t denote the number of customers who visit the shop during a period of time of length t (measured in hours).

(a) Write down the expression for $P(X_t = n)$, where n is a natural number.

(b) Starting from $t_0 = 0$, suppose the first customer arrives at time T, find the cdf and pdf of T.

(c) Based on your result in (b), what distribution does T follow?

For any $t \geqslant 0$, the cdf of T, $F(t)$, satisfies $F(t) = P(T \leqslant t)$. Note that $\{T \leqslant t\}$ means that some customer(s) have/has already visited the shop during the period $[0, t]$, which is equivalent to the event that $\{X_t > 0\}$. As a result,

$$P(T \leqslant t) = P(X_t > 0) = 1 - P(X_t = 0) = 1 - \mathrm{e}^{-\alpha t} \cdot \frac{(\alpha t)^0}{0!} = 1 - \mathrm{e}^{-\alpha t}.$$

So the cdf of T

$$F(t) = \begin{cases} 0, & \text{if } t < 0 \\ 1 - \mathrm{e}^{-\alpha t}, & \text{if } t \geqslant 0 \end{cases}.$$

Using the property that the pdf $f(t) = F'(t)$, we obtain

$$f(t) = \begin{cases} 0, & \text{if } t < 0 \\ \alpha \mathrm{e}^{-\alpha t}, & \text{if } t \geqslant 0 \end{cases}.$$